

(a) sql2skeleton Transformation by SQL Complexity

Easy

How many cars have more than 4 cylinders?
SQL
SELECT COUNT(*)
FROM cars_data
WHERE cylinders > 4
tip values

Skeleton
SELECT count(*)
FROM cars_data
WHERE cylinders > VALUE

Medium

For each stadium, how many concerts?
SQL
SELECT T2.name, COUNT(*)
FROM concert T1 JOIN stadium T2
ON T1.stadium_id = T2.stadium_id
GROUP BY T1.stadium_id
values

Skeleton
SELECT T2.name, count(*)
FROM concert T1 JOIN stadium T2
ON T1.stadium_id = T2.stadium_id
GROUP BY T1.stadium_id

Hard

Countries in Europe with ≥ 3 car makers?
SQL
SELECT T1.country_name FROM countries T1
JOIN continents T2 ON T1.continent=T2.cont_id
JOIN car_makers T3 ON T1.country_id=T3.country_id
WHERE T2.continent = 'Europe'
GROUP BY T1.country_name
HAVING COUNT(*) ≥ 3
values

Skeleton
SELECT T1.country_name FROM countries T1
JOIN continents T2 ON T1.continent=T2.cont_id
JOIN car_makers T3 ON T1.country_id=T3.country_id
WHERE T2.continent = VALUE
GROUP BY T1.country_name
HAVING count(*) ≥ VALUE

Extra Hard

Avg life expectancy, English not official?
SQL
SELECT AVG(life expectancy) FROM country
WHERE name NOT IN (
SELECT T1.name FROM country T1
JOIN country_language T2
ON T1.code = T2.country_code
WHERE T2.language = 'English'
AND T2.is_official = 1)
value

Skeleton
SELECT avg(life expectancy) FROM country
WHERE name NOT IN (
SELECT T1.name FROM country T1
JOIN country_language T2
ON T1.code = T2.country_code
WHERE T2.language = VALUE
AND T2.is_official = VALUE)

(b) Few-Shot Retrieval by Skeleton Similarity

Test question skeleton
SELECT count(*) FROM _ T1
JOIN _ T2 ON ... GROUP BY _
cosine sim (skeletons)

sim = 0.96 ✓ TOP-1
SELECT count(*) FROM _ T1
JOIN _ T2 ON ... GROUP BY _

sim = 0.88 ✓ TOP-2
SELECT count(*) FROM _
JOIN _ ON ... GROUP BY _ ORDER BY _

sim = 0.41 ✗ skipped
SELECT avg(_) FROM _
WHERE _ NOT IN (SELECT ...)

Why it helps:

- ✓ Hides column names → fewer no_such_column errors
- ✓ Structural match → identical JOIN/HAVING pattern
- ✓ Value-agnostic → works across databases